Master 2 Informatique

Data sciences, infrastructure cloud et sécurité

Machine Learning By Practice

Fabrice LAURI

Maître de conférences

à l'Université de Technologie de Belfort-Montbéliard

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- Dimensionality Reduction
- Reinforcement Learning
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Detailed Overview

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- Types of Machine Learning

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Detailed Overview

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Detailed Overview

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Detailed Overview

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Maître de conférences à l'UTBM

CM #1

Introduction

- What is learning?
- Types of Machine Learning
- Reinforcement Learning (standard algorithms)

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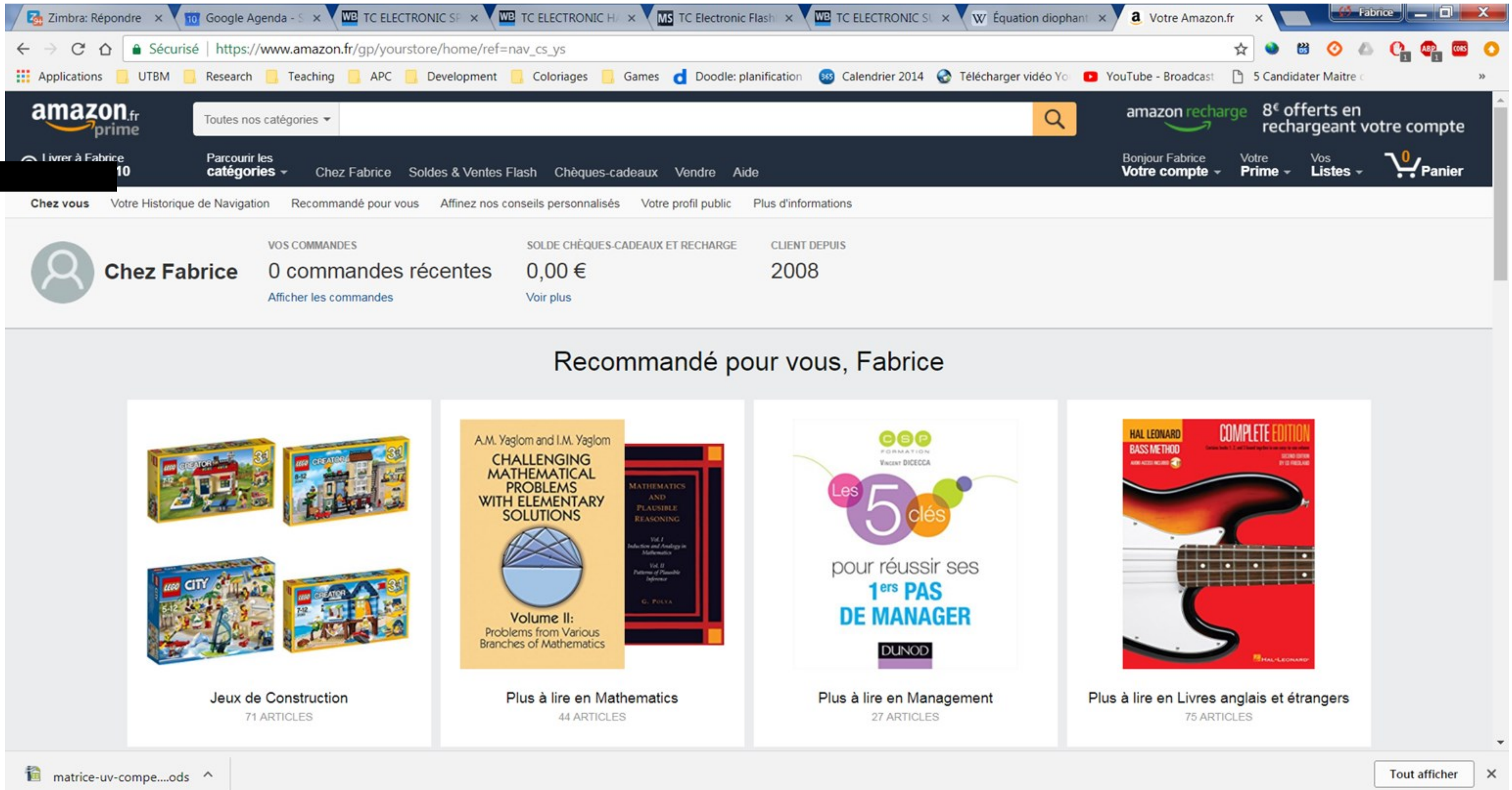
Introduction

Introduction

Suppose you have to develop a website selling all kinds of products.
You want to make it more personalised to the user...



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Introduction

For each person who buys something, you collect some data (country, time, products bought, how paid...).
Based on this data you want to populate a *“Things you might be interested in“* box within your website.
Your hope is that as more people visit your website and more data is collected, you are able to identify trends.
The problem you have is called ***prediction***: given the data you have, predict what the next person will buy.
The reason that you think it might work is that people who seem to be similar often act similarly...
This is an example of ***supervised learning***: we know what the right answers are for some examples, we give them to a learner so that it finds the mapping between examples and answers.

Introduction

Around the world, computers capture and store terrabytes of data every day.

Apart your owns for storing MP3s and holiday pictures, computers belong to shops, banks, hospitals, scientific laboratories and many more...

Banks: pictures of how people spend their money.

Hospitals: what treatments patients are on for which diseases and how they respond to them.

Engine monitoring systems in cars: record information about the engine in order to detect when it might fail.

The challenge is to do something useful with this data.

This is ***data mining***: extracting useful information from massive datasets.

Some questions:

- Can bank detect credit card fraud quickly?
- Can treatments that don't work as well as expected be identified quickly?

Introduction

The size and complexity of these datasets mean that humans are unable to extract useful information from them.
Even the way that the data is stored works against us.
It is easier for us to visualise data than to see it in a table of values...
But if the data has more than three dimensions, we must project it into fewer dimensions (2 or 3) so that we can focus on important aspects within the dataset.
This is called ***dimensionality reduction***.

Machine Learning is becoming popular because the problems of our human limitations go away if we can make computers do the dirty work for us.

Introduction

In fact you interact with machine learning algorithms every day.

They are used in many software programs:
- spam filters
- voice recognition software, like Siri
- lots of computer games
- anti-skid braking and vehicle stability systems
- automatic number-plate recognition systems
- part of the set of algorithms that decide whether a bank will give you a loan

...

Introduction: What learning is?

But what learning actually is?
What learning from data, learning from experiences, are?

Learning is what gives us flexibility in our life: we can adjust and adapt to new circumstances.
The important parts of animal learning are ***remembering***, ***adapting*** and ***generalising***: recognising that last time we were in this situation (saw this data), we tried out some particular action (gave this output) and it worked (was correct), so we'll try it again, or it didn't worked, so we'll try something different.
Generalising is about recognising similarity between different situations so that things that applied in one place can be used in another.

To sum up: **learning is getting better at some task through practice**.

Introduction: Machine Learning...

Machine learning is about making computers modify or adapt their actions (making predictions or controlling a robot...) so that these actions get more accurate, where accuracy is measured by how well the chosen actions reflect the correct ones.

Imagine you are playing chess, Scrabble, or some other game against a computer...

ML merges ideas from neuroscience, biology, statistics, mathematics, physics, to make computers learn.

When studying ML it is particularly important to know the computational complexity of ML algorithms because we might want to use them on very large datasets.
Two complexities: the complexity of training, and the complexity of applying the trained algorithm.

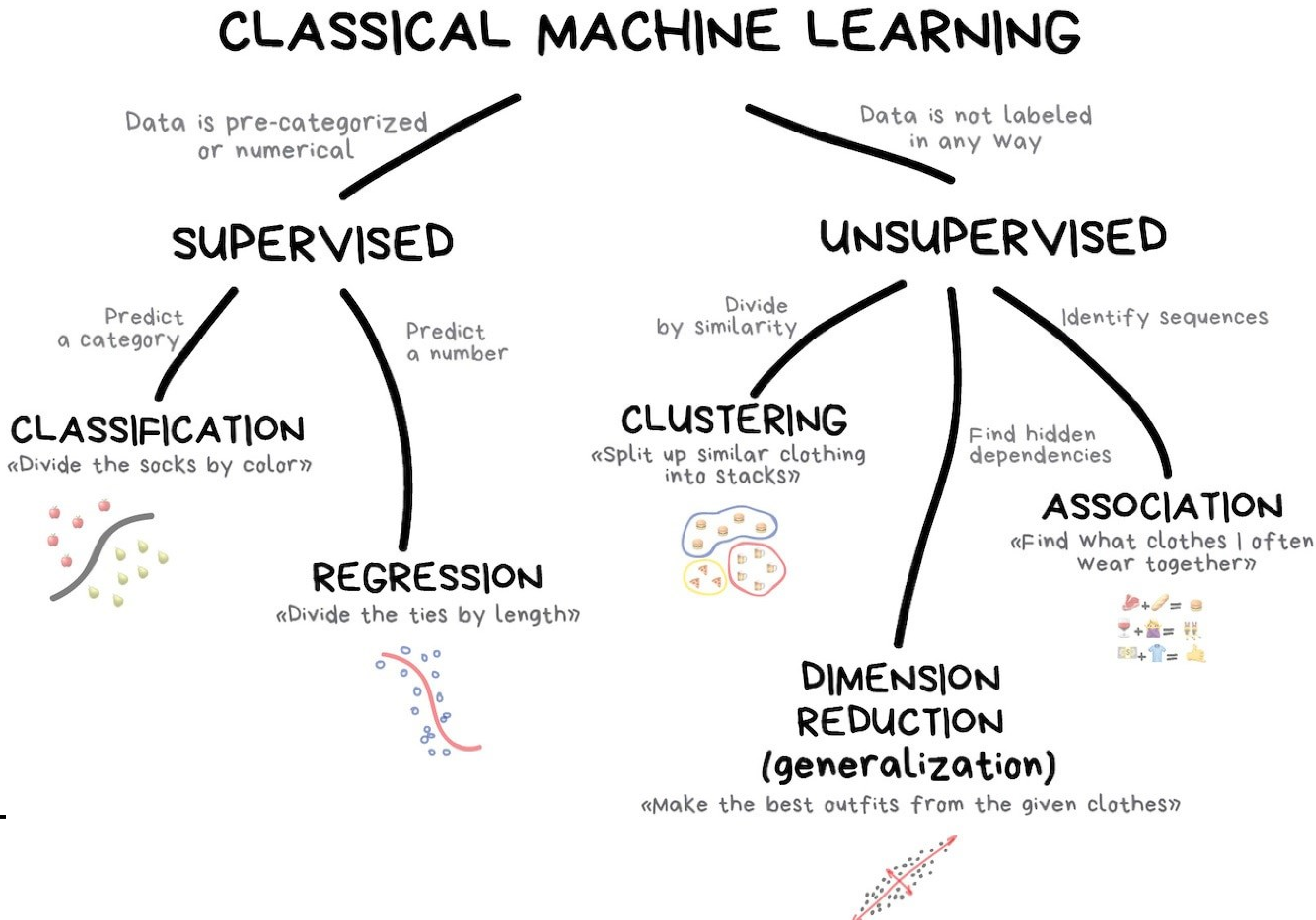
Introduction: Types of Machine Learning

Supervised learning: A training set of examples with the correct responses (targets) are provided and, based on this training set, the algorithm tries to find a function mapping inputs to outputs and that generalises to respond correctly to all possible inputs.

Unsupervised learning: Correct responses are not provided, instead The algorithm tries to identify similarities between the inputs so that inputs that have something in common are categorised together.

Reinforcement learning: RL is the learning a behavior, a sequence of actions. The algorithm gets told a measure of the quality of an action. It has to explore and try out different possibilities until it works out how to correctly perform a given task.
RL is often called learning from trials and errors.

Introduction: Types of Machine Learning



Supervised Learning

The formal supervised learning process involves input variables, which we call X, and an output variable, which we call Y.
We use an algorithm to learn the mapping function from the input to the output. In simple mathematics, the output Y is a dependent variable of input X as illustrated by:

$Y = f(X)$

The end goal is to try to approximate the mapping function f, so that we can predict the output variables Y when we have new input data X.

Examples:

- **Regression:** Predict the price of a house
- **Image classification:** Is it a cat or a dog?
- **Time-Serie Prediction:** How’s the weather today?
- **Text classification:** Who are the unhappy customers? (to predict the sentiment)

Unsupervised Learning

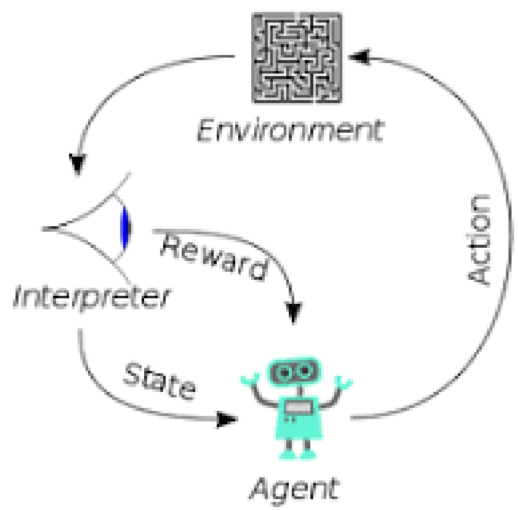
Here we use the data points as references to find meaningful structure and patterns in the observations. Unsupervised learning is commonly used for finding meaningful patterns and groupings inherent in data, extracting generative features, and exploratory purposes.

Examples:

- **Clustering:** Finding customer segments
- **Dimensionality Reduction:** Reducing the complexity of a problem
- **Feature selection**

Reinforcement Learning

Reinforcement learning is a machine learning training method based on rewarding desired behaviors and punishing undesired ones.
In general, a reinforcement learning agent (the entity being trained) is able to perceive and interpret its environment, take actions and learn through trial and error.



Examples:

- training robots that have the ability to grasp various objects
- learning playing games (*Alpha Zero*)
- learning treatments to administer to patients…